

**INTERNSHIP TASK REPORT**  
**ON**  
**SMART TEMPERATURE MONITOR USING**  
**ARDUINO**

**TASK-1**

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# Introduction to Embedded Systems

An **embedded system** is a small, special-purpose computer built into a larger device to do one specific job. Unlike laptops or phones that can run many different programs, an embedded system is designed to perform only one task very well.

It is a combination of **hardware** (electronic parts) and **software** (program) working together inside things like washing machines, cars, or medical devices

## What is an Embedded System?

An embedded system is a microcontroller-based system that performs a specific task in real time, consisting of hardware, application software, and often a Real-Time Operating System (RTOS).

## Applications of Embedded Systems

Embedded systems are used in almost every area of modern life. They make devices smart, automatic, and efficient by performing specific tasks reliably.

**Consumer Electronics:** They run everyday gadgets we use daily, such as smartphones, smartwatches, fitness trackers, and gaming consoles.

**Automotive (Cars):** In cars, they control the engine, safety features, and comfort systems. Examples include anti-lock brakes (ABS), airbags, GPS navigation, and engine control units.

**Healthcare:** They monitor patients and deliver accurate treatment. Common examples are pacemakers, insulin pumps, glucose monitors, and X-ray machines.

**Home Automation:** They make homes smart and energy-efficient. This includes smart thermostats, automatic lights, security cameras, and smart locks.

**Industrial Applications:** Factories use them to control machines and improve production. Examples include robot arms, assembly line controllers, and inventory management systems.

**Aerospace & Aviation:** They help fly planes and spacecraft safely by managing flight control systems, navigation, communication, and satellite operations.

**Robotics:** Embedded systems enable robots to move and make decisions. They are found in manufacturing robots, medical surgery robots, and exploration robots.

**Internet of Things (IoT):** They connect objects to the internet for smart homes, smart cities, connected appliances, and weather sensors

## Microcontroller vs Microprocessor

### Microprocessor

A microprocessor is just the CPU (Central Processing Unit). It needs external chips for memory (RAM, ROM) and input/output (I/O) devices, so the circuit is more complex. It is powerful but uses more power and is not efficient for small tasks.

- Used in: Personal computers, laptops, smartphones
- Pros: Very high processing power, handles complex tasks
- Cons: Needs extra components, larger circuit, higher power consumption

### Microcontroller

A microcontroller is a complete computer on one chip. It has the CPU, memory (RAM, ROM), and I/O ports all built-in, so the circuit is simple and compact. It is efficient, low-power, and perfect for controlled tasks.

- Used in: Washing machines, air conditioners, microwaves, cars
- Pros: Small, cheap, low power, simple circuit
- Cons: Less processing power than microprocessors

| Feature            | Microprocessor                             | Microcontroller                                    |
|--------------------|--------------------------------------------|----------------------------------------------------|
| What it is         | Just the CPU (needs external memory & I/O) | Complete computer on one chip (CPU + memory + I/O) |
| Circuit complexity | Complex (many external connections)        | Simple (everything on chip)                        |
| Power use          | High power consumption                     | Low power consumption                              |
| Cost               | More expensive                             | Less expensive                                     |
| Best for           | Computers run many programs                | Devices that do one specific job                   |

# **Mini Project Description**

## **Title: Smart Temperature Monitor**

### **Project Description**

I made a small project called a Smart Temperature Monitor. It checks the temperature around it all the time. If the temperature gets too high (more than 30°C), it gives an alert. The alert is made by a light (LED) turning on and a buzzer making a sound. This project shows how we can use an Arduino to read sensors and control things automatically.

### **Objectives**

- To make a simple system that checks temperature.
- To see if the temperature goes above 30°C.
- To make a light and a sound when it is too hot.
- To learn how sensors and Arduino work together.
- To keep room or machine temperature safe.

### **Components Required**

- Arduino Uno board
- Temperature Sensor (TMP36 or LM35)
- LED (any colour)
- Small Buzzer
- Resistor (220Ω)
- Breadboard
- Jumper wires (connectors)

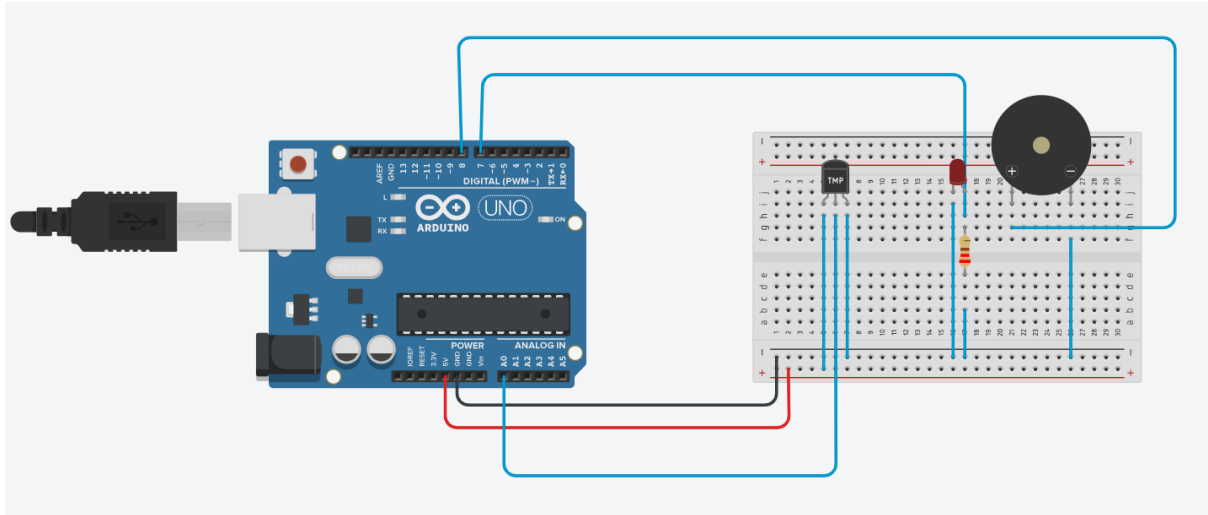
### **Working Principle**

The temperature sensor feels the heat around it and sends a signal to the Arduino. The Arduino reads this signal and changes it into a temperature number. Then, the Arduino checks this number.

- If the temperature is **below 30°C**, nothing happens. The light is off and the buzzer is silent.
- If the temperature is **above 30°C**, the Arduino turns on the light and makes the buzzer beep. This tells us that it is getting too hot.

The temperature number is also shown on the computer screen (Serial Monitor) every second.

## Circuit Diagram



The circuit is very simple:

1. The sensor is connected to the Arduino to read temperature.
2. The LED and buzzer are connected to the Arduino to give the alert.
3. All parts share a common ground (negative wire).

## Connections

### TMP36 Temperature Sensor:

- Left pin → Connect to **5V** on Arduino
- Middle pin → Connect to **A0** (Analog pin) on Arduino
- Right pin → Connect to **GND** on Arduino

### LED:

- Long leg (Positive) → Connect to **Pin 7** through a 220Ω resistor
- Short leg (Negative) → Connect to **GND** on Arduino

### **Buzzer:**

- Positive (+) leg → Connect to **Pin 8** on Arduino
- Negative (–) leg → Connect to **GND** on Arduino

### **Arduino Source Code**

```
int sensorPin = A0;    // Sensor connected to pin A0
int ledPin = 7;        // LED connected to pin 7
int buzzerPin = 8;     // Buzzer connected to pin 8
float temperature = 0; // Variable to store temperature

void setup()
{
    pinMode(ledPin, OUTPUT); // Set LED as output
    pinMode(buzzerPin, OUTPUT); // Set Buzzer as output
    Serial.begin(9600);      // Start communication with computer
}

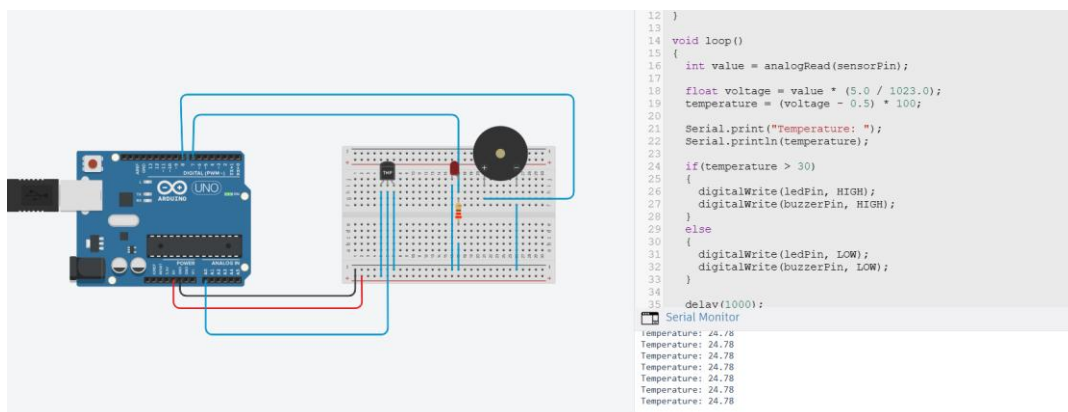
void loop()
{
    int value = analogRead(sensorPin); // Read sensor value
    // Convert value to voltage
    float voltage = value * (5.0 / 1023.0);
    // Convert voltage to temperature in Celsius
    temperature = (voltage - 0.5) * 100;
    // Show temperature on computer
    Serial.print("Temperature: ");
    Serial.println(temperature);
    // Check if temperature is too high
    if(temperature > 30)
    {
        digitalWrite(ledPin, HIGH); // Turn LED ON
        digitalWrite(buzzerPin, HIGH); // Turn Buzzer ON
    }
}
```

else

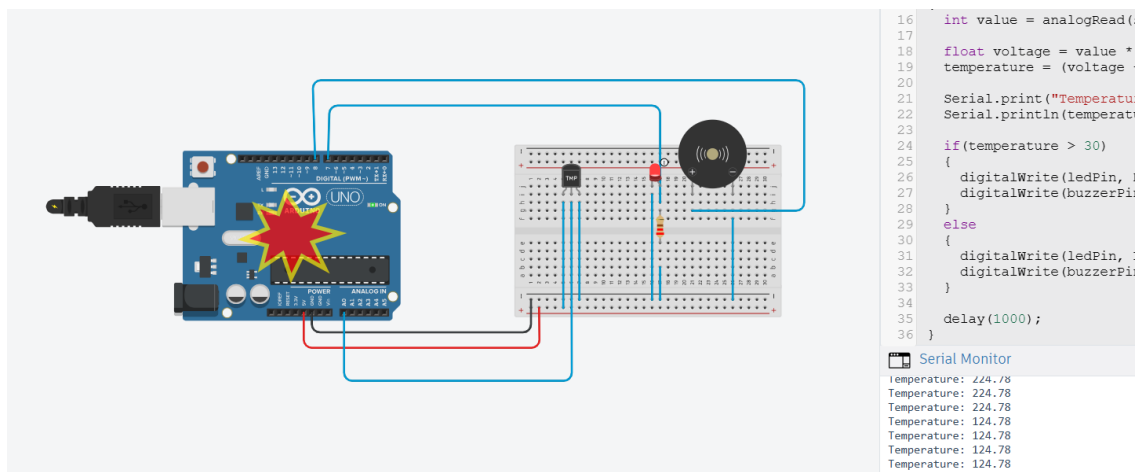
```
{  
  digitalWrite(ledPin, LOW);    // Turn LED OFF  
  digitalWrite(buzzerPin, LOW); // Turn Buzzer OFF  
}  
  
delay(1000); // Wait for 1 second before reading again  
}
```

## Output

Within the threshold value



Beyond threshold value



## Applications

This project can be used in many small ways:

- Checking room temperature in a house.
- Making sure a computer server room does not get too hot.

- Warning if a machine is overheating in a factory.
- Giving a simple fire warning if it gets very hot.

## Conclusion

I successfully made a Smart Temperature Monitor using Arduino. The system works perfectly. It reads the temperature and gives an alert when it gets above 30°C. This project helped me understand how sensors, microcontrollers, and alerts work together in real life. It is a simple but useful example of an embedded system.

## Real-World Applications

In the real world, similar systems are used in:

1. **Homes:** To keep rooms at a comfortable temperature.
2. **Factories:** To stop machines from breaking due to heat.
3. **Hospitals:** To monitor temperature in medicine storage rooms.
4. **Computers:** To prevent CPUs from overheating.
5. **Fire Safety:** To detect heat early and warn people before a fire starts.